

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

May 2012

CE215 OBJECT ORIENTED PROGRAMMING

Date: 05.05.2012, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Which three models are used to describe a system in object oriented modeling? [03]
(b) Compare object oriented approach with procedure oriented approach. [04]
- Q - 2 (a) Explain functional model and role of DFD. [04]
(b) Write a scenario to call your friend using your mobile. His mobile number is already stored in your mobile. [05]
(c) Draw the class diagram for Inventory Management System. [05]

OR

- Q - 2 (a) Explain the concepts of classes, objects and attributes. [04]
(b) Which are the stages of OMT? [05]
(c) Draw the data flow diagram for Library Management System. [05]
- Q - 3 (a) Explain state, events and activity in a state diagram. [04]
(b) Draw the use case diagram for Hospital Management System. [05]
(c) Write a short note on computer graphics. [05]

OR

- Q - 3 (a) Explain (1) Aggregation vs Association (2) Aggregation vs Generalization. [04]
(b) Draw the use case diagram for Hotel Management System. [05]
(c) Write a short note on basic functionality of Matlab. [05]

SECTION – II

Q - 4 Do as directed.

[07]

- (1) Which operators cannot be overloaded in C++?
- (2) Which operators are used with *cin* and *cout* in C++?
- (3) Define Data Encapsulation.
- (4) What is polymorphism?
- (5) A function that doesn't return anything has return type_____.
- (6) *endl* is an operator which is used for newline. State True or False.
- (7) What is the output of the following:

```
main()
{   for (int i=1, j=0; i<5; i++)
    cout<<i<<j;
    return 0;}
```

Q - 5 (a) Write a C++ program to evaluate the following series:

[06]

$1 + 3 + 5 + \dots + n$, where n is input from the user.

- (b) Define a class distance having private data members: meter and cm. Define member functions to input and display the data. Also define an overload operator + adds two objects and assign answer in third object. Write main() function to test the class. **[08]**

OR

Q - 5 (a) Write a C++ program to find whether the entered number is prime or not.

[06]

- (b) Define a class time having members: min and sec. Define member functions to get and print the data. Write a member function that compare two objects and display whichever the larger object. Also define main() to test the class. **[08]**

Q - 6 (a) Draw diagrams that show Multiple & Multilevel inheritance.

[04]

(b) Which are the characteristics of the constructor of a class? **[05]**

(c) Explain public, private and protected access specifiers in detail. **[05]**

OR

Q - 6 (a) Which are the characteristics of the friend function in C++?

[04]

(b) Explain default arguments with example. **[05]**

(c) Explain the need of the virtual base class in inheritance with example. **[05]**
